

Speech Technology



Nguyen Thi Thu Trang

About me

- Engineer & Master in CS, HUST, Vietnam
- PhD in CS, Université Paris-Sud 11, France
- Senior Lecturer and Researcher, SoICT, HUST
 - Software Engineering, Speech Language Processing, Natural Language Processing
 - <https://dblp.org/pid/140/2812.html>
 - <https://scholar.google.com/citations?user=nTljOQYAAAAJ>
- National Golden Globe Award in ICT (2009)
- Founder of Vbee (vbee.ai, vbee.vn)
 - Highest Award at Vietnam Talent for Text-To-Speech (2018)
 - Winner at Grab Venture Ignite for Conversational AI (2020)
 - Winner at Qualcomm Innovation Challenge for on-device Conversational AI (2024)
 - and many more...



0. Course Introduction



Nguyen Thi Thu Trang

Course Introduction

1. Acoustics & Phonetics
2. Speech Dataset
3. Speech Applications

Course Introduction

1. Acoustics & Phonetics

- Sound Representation. Prosody, Intonation, and Accents
- Phonetics Basics. Vietnamese phonology and phonetics
- Signal Representation, Feature Extraction
- Noise Reduction and Silence Removal
- Speech Augmentation
- Normalization and Feature Scaling

2. Speech Dataset

3. Speech Applications

Course Introduction

1. Acoustics & Phonetics

2. **Speech Dataset**

- Dataset Acquisition
- Label Alignment
- Dataset Preprocessing
- Dataset Augmentation
- Dataset Evaluation
- Large Audio-Text Datasets

3. Speech Applications

Course Introduction

1. Acoustics & Phonetics

2. Speech Dataset

3. Speech Applications

- Model Evaluation Metrics
- Model Debugging and Optimization
- Text-to-Speech (TTS)
- Voice Cloning
- Automatic Speech Recognition
- Speech Profiling